INCH-POUND

MIL-DTL-32641(AR)
04 May 2020

DETAIL SPECIFICATION

DUMMY BOOSTER, DEMOLITION CHARGE: NON-ELECTRIC, 30 FT INERT DETONATING CORD - M652, LOAD, ASSEMBLE, AND PACK

This specification is approved for use by US Army Combat Capability Development Command – Armaments Center (CCDC - AC) and is available for use by all Departments and Agencies of the Department of Defense within the distribution limitations noted at the bottom of the page.

1. SCOPE

1.1 Scope. This specification describes the requirements and verification methods for the following items:

DUMMY BOOSTER, DEMOLITION CHARGE: NON-ELECTRIC, 30 FT INERT DETONATING CORD - M652

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army CCDC, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to usarmy.pica.ccdc-ac.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC: 1375

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERICAL ITEM DESCRIPTIONS

A-A-208	-	Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)
A-A-59383	-	Marking Media, Opaque (Porous and Non-Porous Surfaces)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-17124	-	Cord, Detonating and Dummy Cord, Detonating

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Products

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US ARMY CCDC - AC DRAWINGS

8830953	-	Cup
13026123	-	J-hook
13026124	-	Bushing
13038431	-	Dummy Booster, Demolition Charge: Non-Electric, 30 ft Inert Detonating Cord, M652
13042072	-	Cord, Detonating
13042834	-	Cap, Dummy
13096806	-	Combination of Adopted Items for: Dummy Booster, Demolition Charge: Non-Electric, 30 ft Inert Detonating Cord, Packed in Box, Fiberboard
13096807	-	Packing and Marking for Box, Fiberboard for Dummy Booster, Demolition Charge, Inert, M652

(Copies of the above drawings are available by email at usarmy.pica.ccdc-ac.list.drawing-request-help-desk@mail.mil.)

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US ARMY DEFENSE AMMUNITION CENTER (DAC) DRAWINGS

19-48-4116/127N	-	Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets - Dummy Booster, Demolition Charge: Non-Electric, 30 FT Inert Detonating Cord, M652, Packed 120 Per Fiberboard Box, Unitized 24 Boxes Per 40" X 48" Pallet; Approx Box Size 24- 1/2" L X 12-1/4" W X 10-3/4" H

(Copies of these documents may be requested online at https://www.dau.edu/cop/ammo/Pages/Default.aspx.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.1), a sample shall be subjected to the first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Materials, components, and assemblies. All materials, components and assemblies shall conform to Drawing 13038431 and its subsidiary drawings.

3.3 Attachment integrities. The dummy blasting cap and environmental cup shall not separate from the dummy detonating cord after exposure to not less than fifteen pounds (15 lbs) axial force for not less than five (5) seconds.

3.4 Tensile integrity. The detonating cord shall show no evidence of damage after exposure to not less than twenty five pounds (25 lbs) axial force for not less than five (5) seconds.

3.5 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

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3.6 Workmanship. All parts, sub-assemblies, and assemblies shall be clean and free of burrs, chips, sharp edges, cracks, unblended radii, burn marks, checks, sinks, crazes, blisters, excess flash, dirt, grease, oil, rust, corrosion, and other foreign matter.

4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix.

Method of Verification 1 – Analysis 3 – Examination 2 – Demonstration 4 – Test		Classes of Verification A – First article B – Conformance						
Section 3 and 5 Requirements	Description	Verification Method				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article inspection			X	X	X		4.2
3.1.2	Conformance inspection			X	X		X	4.3
3.2	Materials, components, and assemblies			X	X	X	X	4.3.4.1 - 4.3.4.5
3.3	Attachment integrities				X	X	X	4.4.1
3.4	Tensile integrity				X	X	X	4.4.2
3.5	Ammunition lot numbering			X		X	X	4.3.3
3.6	Workmanship			X		X	X	4.3.4.1 - 4.3.4.8
5.1	Packaging and marking			X	X	X	X	4.3.4.6 - 4.3.4.8
5.1.1	Marking ink/media			X		X	X	4.3.4.5, 4.3.4.7, 4.3.4.8

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified, a sample shall be subjected to first article inspection in accordance with Table II. First article samples shall be manufactured using the same vendors, methods, materials, equipment, processes, procedures and facilities as will be used during regular production.

4.2.1 First article rejection. If one (1) or more assemblies, subassemblies, materials, or components fail to comply with the first article requirements, the sample(s) and first article shall be rejected.

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TABLE II. First article tests and inspections.

Examination or test	Drawing(s)	Sample Size	Requirement Paragraph	Paragraph Reference / Inspection Method
Cup	8830953	30	3.2	4.3.4.1
J-hook	13026123	30	3.2	4.3.4.2
Cap, dummy	13042834	30	3.2	4.3.4.3
Bushing	13026124	30	3.2	4.3.4.4
Dummy detonating cord	13042072	MIL-DTL-17124	3.2	MIL-DTL-17124
M652 assembly	13038431	1/		
Examination for defects		30	3.2	4.3.4.5
Attachment integrities		30	3.3	4.4.1
Tensile integrity		30	3.4	4.4.2
Fiberboard box	13096807	2/	5.1	4.3.4.7

Notes:
1/ Sequentially subject 30 samples to the following tests, in the order specified: examination for defects, attachment integrities, and tensile integrity.
2/ The examinations shall be performed on one (1) empty pack.

4.3 Conformance inspection.

4.3.1 Conformance rejection. If any assembly, subassembly, material, or component fails to comply with the conformance inspection requirements of 4.3 and its subparagraphs, the lot from which the sample was taken shall be rejected.

4.3.2 Inspection lot formation. Inspection lots shall conform to the requirements of MIL-STD-1916. Inspection lots shall contain materials from not more than one lot interfix number from one manufacturer (when applicable). When not applicable, lot formation shall be homogeneous. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings and revisions, and the same specifications and revisions.

4.3.3 Ammunition lot number. Visually verify that an ammunition lot number has been assigned to each lot as described in MIL-STD-1168.

4.3.4 Inspection requirements by classification of characteristics.

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a. For the conformance inspection paragraphs 4.3.4.1 through 4.3.4.8, the definitions of critical, major and minor defects are provided in paragraph "Definitions" of MIL-STD-1916.
b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.
c. Alternative conformance acceptance. Alternate conformance procedures may be proposed (See paragraph "Acceptance by contractor-proposed provisions" of MIL-STD-1916).

4.3.4.1 Cup.			Drawing Number 8830953-2 Next Higher Assembly 13042834	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Cup split or cracked	Level II	3.2	Visual
102	Outside diameter	Level II	3.2	Gage
Minor				
203	Workmanship	Level I	3.6	Visual

4.3.4.2 J-hook.			Drawing Number 13026123-4 Next Higher Assembly 13038431	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Lengthwise hole diameter	Level II	3.2	Gage
102	Type IE detonating cord not accepted and retained	Level II	3.2	Manual
Minor				
201	Workmanship	Level I	3.6	Visual

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4.3.4.3 Cap, dummy.			Drawing Number 13042834 Next Higher Assembly 13038431	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Diameter of holes	Level II	3.2	Gage
102	Location of holes	Level II	3.2	Gage
103	Orientation of holes	Level II	3.2	Visual
Minor				
201	Color (general match) incorrect	Level I	3.2	Visual
202	Coating incomplete	Level I	3.2	Visual
203	Length	Level I	3.2	Gage
204	Workmanship	Level I	3.6	Visual

4.3.4.4 Bushing.			Drawing Number 13026124 Next Higher Assembly 13038431	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Length	Level II	3.2	Gage
102	Outside diameter	Level II	3.2	Gage
103	Inside diameter	Level II	3.2	Gage
Minor				
201	Workmanship	Level I	3.6	Visual

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4.3.4.5 Dummy booster, demolition charge: non-electric, 30 ft inert detonating cord, M652.			Drawing Number 13038431 Next Higher Assembly 13096807	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Holes not present or obstructed	100%	3.2	Visual
102	Components not present	Level II	3.2	Visual
103	Cup dented, split, or cracked	Level II	3.2	Visual
104	Crimp diameters (x2)	Level II	3.2	Gage
105	Crimp widths (x2)	Level II	3.2	Gage
106	Location of 1st crimp from cup edge (x2)	Level II	3.2	Gage
107	Location of 2nd crimp from 1st crimp (x2)	Level II	3.2	Gage
108	Location of pentagonal flags	Level II	3.2	Gage
109	Detonating cord	MIL-DTL-17124	3.2	MIL-DTL-17124
110	Assembly length	10 1/	3.2	Gage
111	Attachment integrities	10 1/	3.3	4.4.1
112	Tensile integrity	10 1/	3.4	4.4.2
Minor				
201	Markings missing, illegible or incorrect	Level I	5.1.1	Visual
202	Pentagonal flag markings missing, illegible, or incorrect	Level I	3.2	Visual
203	Workmanship	Level I	3.6	Visual

Notes:
1/ Accept on zero (0) failures; reject on one (1) or more failures. Sequential testing shall be performed in the following order: assembly length, attachment integrities, then tensile integrity. All samples may be returned to the lot after successful test completion and approved rework.

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4.3.4.6 Packaging and marking drawing for the fiberboard box (unsealed).			Drawing Number 13096807	
			Next Higher Assembly 19-48-4116/127N	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Missing items	Level II	5.1	Visual
Minor				
201	Workmanship	Level I	3.6	Visual

4.3.4.7 Packaging and marking drawing for the fiberboard box (sealed).			Drawing Number 13096807	
			Next Higher Assembly 19-48-4116/127N	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Sealing strip torn, badly wrinkled or otherwise fails to seal box completely or correctly	Level II	5.1	Visual
102	Markings missing, incorrect, illegible, incorrect size/height, or not in proper location	Level II	5.1.1	Visual / Gage
Minor				
201	Workmanship	Level I	3.6	Visual

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4.3.4.8 Unitization.			Drawing Number 19-48-4116/127N Next Higher Assembly	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Wood preservative and heat treatment markings for the pallet missing, incorrect, illegible, or not in proper location	VL II	5.1.1	Visual
102	Pallet load markings -including 2D bar codes - missing, incorrect, illegible, or not in proper location	VL II	5.1.1	Visual
103	Filler assembly not built, treated, or marked correctly	VL II	5.1	Visual
104	Boxes incorrectly oriented or positioned on pallet	VL II	5.1	Visual
105	Strapping not applied correctly or incorrect type	VL II	5.1	Visual
106	Unit load not tightly strapped or any strap loose	VL II	5.1	Visual / Manual
Minor				
201	Workmanship	Level I	3.6	Visual

4.4 Test methods.

4.4.1 Attachment integrities. Secure the dummy blasting cap. Apply the specified axial load to the dummy detonating cord for the specified length of time. Verify that the dummy blasting cap remains attached to the dummy detonating cord. Unsecure the dummy cap. Secure the environmental cup. Apply the specified axial load to the dummy detonating cord for the specified length of time. Verify that the environmental cup remains attached to the dummy detonating cord. The sample(s) and lot shall be rejected if there is separation of the blasting cap or the environmental cup.

4.4.2 Tensile integrity. Secure a portion of dummy detonating cord not less than twenty four (24) inches from the dummy blasting cap and not less than twelve (12) inches in length. Apply the specified axial force to the dummy detonating cord for the specified length of time. The sample(s) and lot shall be rejected if any of the dummy detonating cord separates.

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5. PACKAGING

5.1 Packaging and marking. The materials, components, and assemblies shall comply with all the requirements of Drawing 13096806 and all its subsidiary drawings.

5.1.1 Marking ink/media. The ink and/or marking media shall be in accordance with A-A-208 and/or A-A-59383. All markings shall be smooth, uniform, clear in appearance and free of conspicuously ragged edges. There shall be no bleeding of color markings or the plastic exterior to the extent that there is a mottled or streaky appearance. The markings shall show no lifting, wrinkling, or other visible effect on the plastic's surface. The markings shall be applied on clean, dry surfaces that shall be free of dirt, oil, grease, water, corrosion, or other foreign matter. All required markings shall be present, neat, legible, only where specified, and sharply defined.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M652 (M152 Inert) is used for training the M151 and M152.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, and of all reference documents cited in Section 2 and applicable documents from Section 6.
b. Requirement/CDRL for First article inspection.
c. Requirement/CDRL for conformance inspection.
d. Requirement/CDRL for Ammunition Data Cards (ADC) as specified in MIL-STD-1168.
e. Requirement/CDRL for Acceptance Inspection Equipment (AIE) and associated test procedures.
f. Requirement/CDRL for Alternate Acceptance Plans.
g. Packaging requirements. Special requirements exist for the packaging of this item. Detailed directions and additional information may be obtained from FCDD-ACE-LP.

6.3 Document history. This document supersedes PRF14000035C Performance Specification for the Booster, Demolition Charge, Practice: Inert, M152.

6.4 Subject term (key word) listing.
Underground
Quarry operations
Staged demolitions
MDI (Modernized Demolition Initiators)

6.5 Overview of changes. The following approved Engineering Change Proposal (ECP) is being incorporated into this document: R20Q2002.

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